MICROWAVE BLUETOOTH NLC SENSOR

MSM-HT0/09P3/NLC

Accessory | Sensor | Outdoor | Bi-Level | Microwave | 3-Pin

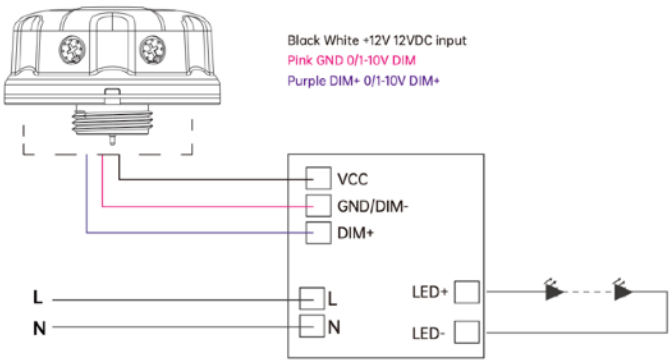
Compatible Controller: RM-HT/05S

Input	Input Range	12Vdc
	Voltage Range	10-15Vdc
	Current	Remote Controlled: <30mA
Output	Signal	DIM 0-10V
	Connection	TipDIM+, RingVCC
	Standby Power	<0.5W
blueⒶRCH Bluetooth	Bluetooth Wireless Range	>65ft
	APP Download	Supported on Apple App Store
Environment	Working Temp	-13°F to 140°F
	Storage Temp	-40°F to 176°F | Humidity: 85% (non-condensation)
Microwave	Frequency	5.8GHz ± 75MHz
	Microwave Power	<0.3mW
	Installation Height	4m/13.12 Min.
	Detection Distance	≥3m/9ft
	Detection Angle	30-150° (No Lensing)
Sensor Parameter	Detection Area	25%/50%/75%/100%
	Holdtime	Remote control Options: 5s/ 30s/ 1min/ 3min/ 5min/ 10min/ 20min/ 30min
	Daylight Threshold	Disable/400Lux/350Lux/300Lux/250Lux/200Lux/ 120Lux/ 80Lux/50Lux/30Lux/10Lux/2Lux
	Standby Dimming Level	10%/20%/30%/50%
	Standby Period	Remote control Options: 0s/ 10s/ 30s/ 1min/ 5min/ 10min/ 30min/ 60min/+∞
	Dusk/Dawn Sensing/Photocell	Daylight threshold as 30lux/ 50lux/ 80lux/ 120lux/ 200Lux/ 250Lux/300Lux/ 350Lux/ 400Lux Standby period as +∞ ; Standby dimming level as 10%/20%/30%
	Daylight Harvesting	1. Adjust "daylight" value higher than 50lux 2. Preset "standby period" 0S 3. Press MW/PIR button 3 times till MW/PIR icons both flashing on LCD screen, daylight harvesting function enabled.
Certification	Environmental Requirements	In accordance with CE RoHS
	IP Rating	IP65

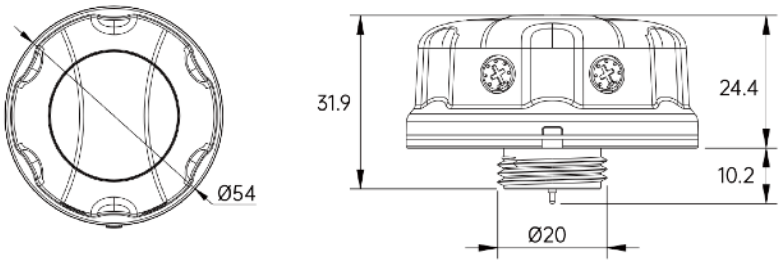
APPLICATION

MSM-HTO/09P3/NLC can be integrated with a lighting fixture to provide multiple occupancy sensing control schemes with multi-level or continuous dimming by sensing the presence and movements of the occupant and the daylight available in the space. A basic wiring diagram is included at next page for reference.

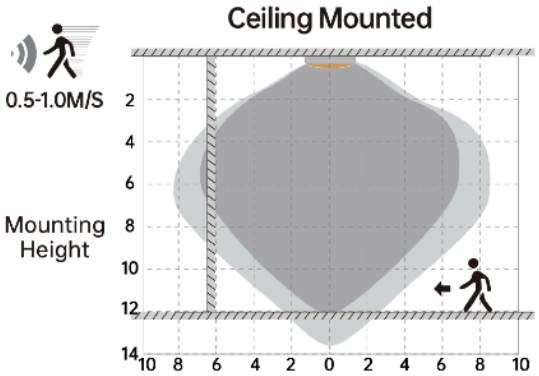
WIRING DIAGRAM

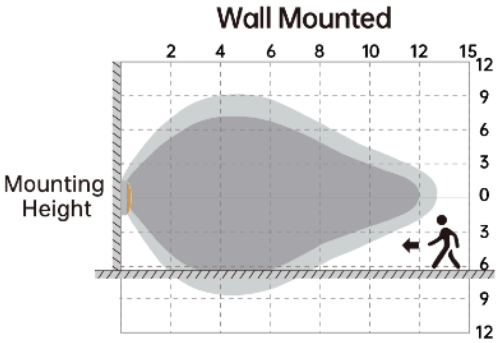


DIMENSIONS



DETECTION RANGE





Highest mounting height is 50FT

This figure indicates the maximum distance at the highest mounting height with 100% sensitivity.

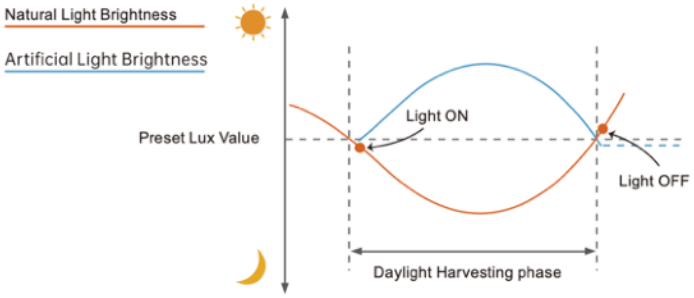
APPLICATION

1. Suitable for indoor applications and ceiling mount installation. Adjust sensitivity properly if installed on side wall due to sensitivity.
2. Adjust sensitivity properly when the sensor is applied in small/narrow/metal-built/metal spaces
3. Microwave sensor can't be placed under or inside metal
4. Microwave module must directly face the detection area with edge lower than light fixture.
5. Keep the sensor away from vibration equipment, air-conditioning outlets, smoke extractors and like conditions to avoid unwanted trigger
6. Keep the sensor module away from AC input and DC output to avoid high/low frequency signal interference.
7. Keep at least 2m/6.5ft distance between microwave sensors; 1.5m/4.9ft between the sensor and other wireless devices such as routers to avoid possible radio interference
8. Daylight testing delivered in bright day without shadow or specially designed lampshade or lens
9. Dimming performance differs when connected to different drivers; If the driver can't completely turn OFF, neither can sensor
10. Input power voltage must be stable with float less than 10%.
11. The first time sensor is powered ON, light will be ON 100% for about 10 seconds then dims to standby level or OFF
12. Distance detection is delivered by testing person about 165cm in open area as reference, the result differs by size and speed of moving objects, mounting height and real-life situation

PERFORMANCE

1. Daylight Harvesting

1. Adjust "daylight" value higher than 50lux
2. Preset "standby period" 0S
3. Press MW/PIR button 3 times till MW/PIR icons both flash on LCD screen, daylight harvesting function enabled.



When ambient brightness is lower than preset lux level, sensor will turn on light automatically and keep dimming according to the change of the ambient brightness; when outside is getting darker, the inside will be brighter, and brighter darker.

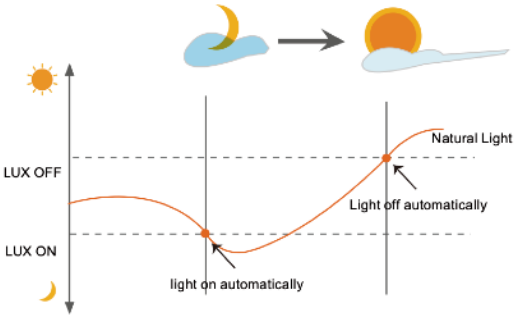
Light OFF when ambient brightness becomes higher than the preset lux level.

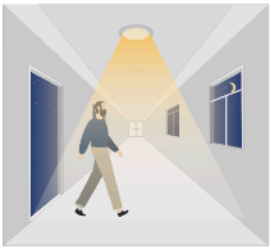
2. Dusk/Dawn

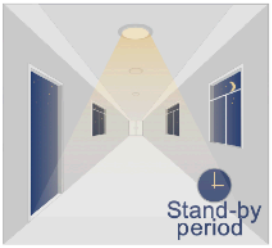
MSM-HTO/07/BLS is able to differentiate artificial light from natural light. It will automatically turn OFF the fixture when ambient brightness exceeds preset lux levels.

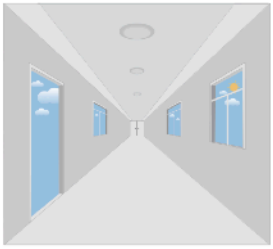
Precondition of Dusk/Dawn Functions:

1. Standby period is +∞
2. Standby dimming level is on 10%, 20% or 30%
3. Daylight threshold is on 30lux / 50lux / 80lux / 120lux / 200lux / 250Lux / 300lux / 350lux / 400lux









With insufficient ambient brightness, sensor turns on light and keeps it at standby dimming level even if there is no motion or presence.

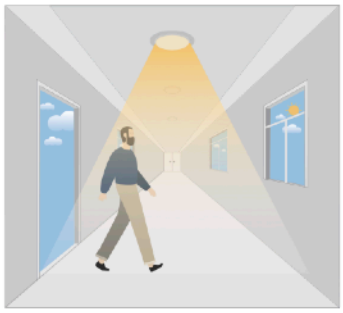
When sensor detects motion or presence it will bring the light level up to 100%.

After motion is no longer detected, fixture remains at 100% for hold time.

After pre-set hold time period it will dim light to standby dimming level again and always keep it.

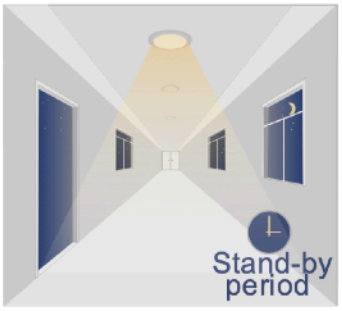
With sufficient ambient brightness, sensor will turn OFF light automatically.

When Daylight Function is Disabled



Sensor turns ON light when motion is detected.

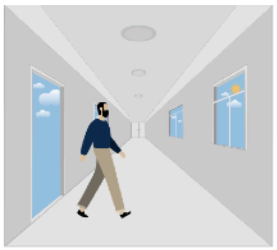
Sensor keeps for a hold time period after motion leaves.



Sensor dims light to standby dimming level after hold time if there is still no motion.

Sensor turns OFF light after standby period.

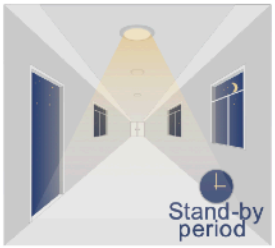
Daylight Threshold Settings



With sufficient daylight, the sensor keeps light OFF even motion gets detected.

With insufficient daylight, the sensor turns light ON when motion gets detected.

After there's no motion detected, the sensor keeps light ON 100% for hold-time.



After hold-time, sensor dims light to standby dimming level for standby period.

If the standby period has been set as 0s, sensor turns light OFF automatically after hold-time.

The sensor turns OFF light automatically after the standby period when there's no motion detected.

ARCHIPELAGO LIGHTING

Archipelago Lighting, Inc. is a leading prime manufacturer of commercial and residential lighting products in USA and China. We are a team of passionate lighting professionals whose goal is to improve life through disruptive LED lighting products. We manufacturer a wide variety of products to support the growing adoption of energy efficient LED.